

Exploratory Data Analysis (EDA)

Titanic Dataset — Code Review, Graph Interpretation, and Knowledge Check

Course: ITAI 1371

Notebook: L04_Group4_ITAI1371.ipynb

Prepared from the uploaded notebook and the corrected code examples discussed in this chat.

Important note: the uploaded notebook did not include saved graph outputs. The graph explanations in this report describe expected patterns for the Titanic dataset; they are not a verification of plots generated by executing this notebook.

1. Purpose of the Lab

Exploratory Data Analysis (EDA) is used to understand a dataset before building a machine-learning model. It helps identify distributions, missing values, unusual values, and relationships between variables. This lab uses Titanic passenger data to explore survival in relation to passenger class, gender, age, embarkation port, and ticket fare.

2. Code Review: Main Issues

Dataset variable mismatch: The notebook has a loading variable written as `URLdf` while later cells refer to `df`. Use `df` consistently.

Commands joined together: Some code lines appear joined or placed after comments. Put each Python statement on its own line so it runs.

Plot commands: Each plotting cell should create the figure, set labels/title, and call `plt.show()` on separate lines.

Incomplete experiment cells: The two experiment cells need code to create the embarkation-port and fare-versus-survival graphs.

3. Corrected Code

Cell 2 — Load the dataset

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(
    "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
)

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
df.info()
```

Cell 4 — Summary statistics

```
print("--- Descriptive Statistics ---")
print(df.describe())

print("--- Key Insights from Statistics ---")
print(f"The average age was {df['Age'].mean():.1f} years.")
print(f"The overall survival rate was {df['Survived'].mean():.1%}.")
print(
    f"Fares ranged from ${df['Fare'].min()} "
    f"to ${df['Fare'].max()}."
)
```

Cell 7 — Survival distribution

```
sns.set_style("whitegrid")

plt.figure(figsize=(8, 6))
sns.countplot(x="Survived", data=df)
plt.title("Survival Distribution (0 = Died, 1 = Survived)")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")
plt.show()

print("More passengers died than survived.")
```

Cell 9 — Survival by passenger class

```
plt.figure(figsize=(10, 6))
sns.countplot(x="Pclass", hue="Survived", data=df)
```

```
plt.title("Survival by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["Died", "Survived"])
plt.show()
```

Cell 11 — Survival by gender

```
plt.figure(figsize=(10, 6))
sns.countplot(x="Sex", hue="Survived", data=df)
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["Died", "Survived"])
plt.show()
```

Cell 13 — Age distribution

```
g = sns.FacetGrid(df, col="Survived", height=5)
g.map(plt.hist, "Age", bins=20)
g.set_axis_labels("Age", "Number of Passengers")
g.set_titles("Survived = {col_name}")
plt.show()
```

Cell 16 — Survival by port of embarkation

```
plt.figure(figsize=(10, 6))
sns.countplot(x="Embarked", hue="Survived", data=df)
plt.title("Survival by Port of Embarkation")
plt.xlabel("Port (C = Cherbourg, Q = Queenstown, S = Southampton)")
plt.ylabel("Number of Passengers")
plt.legend(title="Survived", labels=["Died", "Survived"])
plt.show()
```

Cell 18 — Ticket fare by survival

```
plt.figure(figsize=(8, 6))
sns.boxplot(x="Survived", y="Fare", data=df)
plt.title("Ticket Fare by Survival")
plt.xlabel("Survived (0 = Died, 1 = Survived)")
plt.ylabel("Ticket Fare")
plt.show()
```

4. How to Read the Graphs

Graph 1 — Survival distribution

The x-axis uses 0 for passengers who died and 1 for passengers who survived. The y-axis shows passenger counts. In the commonly used Titanic dataset, more passengers died than survived.

Graph 2 — Survival by passenger class

The x-axis shows first, second, and third class. Bars compare deaths and survivors in each class. First-class passengers generally had a higher survival proportion than third-class passengers. This is an association, not proof that class alone caused the outcome.

Graph 3 — Survival by gender

The x-axis shows male and female passengers, and the bars compare outcomes. Females had a substantially higher survival proportion than males in the dataset. Gender is strongly associated with survival in this historical dataset.

Graph 4 — Age distribution

The histograms compare ages of passengers who died and survived. The plots show the number of passengers in age ranges, not survival rates by age. Age patterns should be interpreted carefully because the group sizes differ and some ages are missing.

Experiment 1 — Survival by port of embarkation

The x-axis uses C (Cherbourg), Q (Queenstown), and S (Southampton). Compare the death and survival bars within each port. The proportions can differ by port, but port is not necessarily the cause; passenger class and other factors may be involved.

Experiment 2 — Ticket fare by survival

A boxplot compares ticket fares for passengers who died (0) and survived (1). The line in each box is the median, the box represents the middle half of values, and points beyond the whiskers may be outliers. Survivors generally paid higher fares, but this does not prove that a higher fare directly caused survival.

5. Knowledge Check — Suggested Answers

1. What is the primary goal of EDA?

EDA helps us understand a dataset before building a machine-learning model. It can reveal patterns, missing values, outliers, and relationships between variables.

2. Based on the plots, what kind of person had the best chance of surviving the Titanic?

Female passengers and passengers traveling in first class had higher survival proportions. Children also show survival patterns worth examining. These are associations observed in the data, not guarantees for individuals.

3. Why visualize data instead of looking only at summary statistics?

Graphs make patterns and relationships easier to see. Summary statistics provide numbers such as averages and counts, while graphs help compare groups, understand distributions, and spot unusual values.

6. Run Checklist

- Run the dataset-loading cell first and confirm that `df` is created.
- Run the remaining cells in order.

- Confirm each plotting cell displays its graph.
- If a dataset download error appears, check the internet connection or share the exact error message.
- Compare the actual graphs with the explanations in this report before submitting.

Limitations: This report provides corrected example code and expected interpretations. The notebook was not executed here, and the original file did not contain saved graph images or outputs. Run the notebook to verify the actual values and visual results.